Oracle Utilities Service and Measurement Data Foundation v26.4HOT

Product Fix Design

Bug 39007114

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Oracle Utilities Service and Measurement Data Foundation Product Fix Design

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Reported Issue

Description

BCS generates bill with multiple headers when there is a rate change within a billing period

When an SA has a Rate change within a billing period, BCS generates bill segment with multiple headers (one for each Rate schedule) in spite of SA Type configuration set up to select the rate at the start of the billing period.

When an SA has multiple rate schedules within the billing period:

- The base algorithm “Derive Usage Group based on Rate & Installed Device” selects multiple Usage Groups based on the rate schedules.
- This results in the creation of multiple Sub UTs.
- Consequently, multiple Bill Calculation headers are generated—one per rate schedule.
- This occurs even though the SA Type is configured to select a single rate, effectively overriding the SA Type configuration during usage calculation.

Steps To Recreate

1. Create V set up for an Interval MC.
2. Ensure algorithm Derive Usage Group Based On Rate and Installed Device (D2-DRVUSGGRP) is plugged-in on the US Type.
3. Create usage subscription
    - Add the SP
    - Rate Schedule is configured as
        - 01-01-2025 → ZZEU1 → Configure Usage Group 1 in Rate Schedule (D2-CCBRateScheduleLookup) extendable lookup.
            - Usage Group 1 with Math Usage Rule
        - 01-03-2025 → ZZEU2 → Configure Usage Group 2 in Rate Schedule (D2-CCBRateScheduleLookup) extendable lookup.
            - Usage Group 2 with Math Usage Rule
    - Ensure interval MC has reads from 01-01-2025 12:00AM to 01-05-2025 12:00AM
4. Create usage transaction for dates
    - Interval Start 01-01-2025 12:00AM
    - Interval End From 01-05-2025 12:00AM
    - Interval End To 01-05-2025 12:00AM
    - Scalar Start 01-01-2025 12:00AM

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- Scalar End From 01-04-2025 12:00AM
- Scalar End To 01-04-2025 11:59PM

5. Two Sub UT's are created
    - Sub UT 1 for the period 01-01-2025 -- 01-03-2025 → Usage Group 1
    - Sub UT 2 for the period 01-03-2025 -- 01-05-2025 → Usage Group 2

Observe that there is no way for the User to say only pick a single effective rate as of start date/time even when there is a rate change within the period.

Solution Design

Summary

When an SA has a rate change within a billing period and the SA Type is configured to select the rate effective at the start of the billing period, BCS should:

- Apply only one rate schedule for the entire bill period
- Generate a single Bill Segment header
- Use a single Usage Group (No Sub UT's created), regardless of mid-period rate changes

Objects modified:

1. Algorithm Type: D2-DRVUSGGRP
    a. Added algorithm soft parm 3 - Use Only One Rate (Y/N)
    b. Updated algorithm soft parm 2 - Suppress Partial Day Effective Rates (Y/N)
    c. Updated Detailed Description
2. Algorithm Type: D2-DRVUSGGRP
    a. Added algorithm soft parm 3 - Use Only One Rate (Y/N)
3. Plug-in script: D2-DrvUsgGrp
    a. Pass the algorithm soft parm 3 - Use Only One Rate (Y/N) to the service script: D2-CreWRL
4. Service script: D2-CreWRL
    a. Updated schema
        <useOnlyOneRate dataType="string"/>
    b. Updated step 20, if the Use Only One Rate is Y, use the first effective rate as of start date/time and terminate for the logic.

Note to customer: Once the SA Type is configured to select a single rate, i.e. SA Type configuration is set up to select the rate at the start of the billing period, User needs to make the corresponding change in MDM for the Algorithm: D2-DRVUSGGRP () to update the soft parm 3 - Use Only One Rate (Y/N) to "Y".

Change Type	Y/N
Does this fix require any Admin (Soft) Table Changes to be made by the client?	N
Does this fix require upgrade data changes to be applied?	N
Does the fix involve DB Changes?	N

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Is there an impact on any user interface or any other on-line behavior?	N
Is there an impact on any batch process?	N
Is there an impact on CM – Custom Modification processing?	N
Does this fix require an application viewer update for Algorithm data (run F1-AVALG)?	N
Does this fix require an application viewer update for MO data (run F1-AVMO)?	N
Does this fix require an application viewer update for Table data (run F1-AVTBL)?	N

Other Changes

Algorithms

Derive Usage Group Based On Rate and Installed Device (D2-DRVUSGGRP) (Change)

Pass the algorithm soft parm 3 - Use Only One Rate (Y/N) to the service script: D2-CreWRL

Service Script

Create Work Rate List (D2-CreWRL) (Change)

if the Use Only One Rate is Y, use the first effective rate as of start date/time and terminate for the logic.

Data Clean-Up / Fixes

Not applicable

Recommended Test Cases

Test Case 1: Sub UT's are created when rate change happens in between the UT period and Use Only One Rate (Y/N) = blank (Default existing behavior)

1. Create V set up for an Interval MC.

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2. Ensure algorithm Derive Usage Group Based On Rate and Installed Device (D2-DRVUSGGRP) is plugged-in on the US Type.
3. Create usage subscription
    - Add the SP
    - Rate Schedule is configured as
        - 01-01-2025 → ZZEU1 → Configure Usage Group 1 in Rate Schedule (D2-CCBRateScheduleLookup) extendable lookup.
            - Usage Group 1 with Math Usage Rule
        - 01-03-2025 → ZZEU2 → Configure Usage Group 2 in Rate Schedule (D2-CCBRateScheduleLookup) extendable lookup.
            - Usage Group 2 with Math Usage Rule
    - Ensure interval MC has reads from 01-01-2025 12:00AM to 01-05-2025 12:00AM
4. Create usage transaction for dates
    - Interval Start 01-01-2025 12:00AM
    - Interval End From 01-05-2025 12:00AM
    - Interval End To 01-05-2025 12:00AM
    - Scalar Start 01-01-2025 12:00AM
    - Scalar End From 01-04-2025 12:00AM
    - Scalar End To 01-04-2025 11:59PM
5. Two Sub UT's are created
    - Sub UT 1 for the period 01-01-2025 -- 01-03-2025 → Usage Group 1
    - Sub UT 2 for the period 01-03-2025 -- 01-05-2025 → Usage Group 2

Test Case 2: No Sub UT's are created when rate change happens in between the UT period and Use Only One Rate (Y/N) = Y (new behavior)

1. Create V set up for an Interval MC.
2. Ensure algorithm Derive Usage Group Based On Rate and Installed Device (D2-DRVUSGGRP) is plugged-in on the US Type.
    a. Algorithm soft parameter "Use Only One Rate (Y/N) = Y"
3. Create usage subscription
    a. Add the SP

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b. Rate Schedule is configured as

i. 01-01-2025 → ZZEU1 → Configure Usage Group 1 in Rate Schedule (D2-CCBRateScheduleLookup) extendable lookup.

1. Usage Group 1 with Math Usage Rule

ii. 01-03-2025 → ZZEU2 → Configure Usage Group 2 in Rate Schedule (D2-CCBRateScheduleLookup) extendable lookup.

1. Usage Group 2 with Math Usage Rule

c. Ensure interval MC has reads from 01-01-2025 12:00AM to 01-05-2025 12:00AM

4. Create usage transaction for dates

a. Interval Start 01-01-2025 12:00AM

b. Interval End From 01-05-2025 12:00AM

c. Interval End To 01-05-2025 12:00AM

d. Scalar Start 01-01-2025 12:00AM

e. Scalar End From 01-04-2025 12:00AM

f. Scalar End To 01-04-2025 11:59PM

5. UT is created (No Sub UT's are created) with the first effective rate as of UT start date/time with Usage Group 1

Test Case 3: No Sub UT created when period is less than 24 hours due to rate change, Suppress Partial Day Effective Rates (Y/N) = Y and Use Only One Rate (Y/N) = Y

1. Create V set up for an Interval MC.
2. Ensure algorithm Derive Usage Group Based On Rate and Installed Device (D2-DRVUSGGRP) is plugged-in on the US Type.
   - Algorithm soft parameter "Use Only One Rate (Y/N) = Y"
   - Suppress Partial Day Effective Rates (Y/N) = Y
3. Create usage subscription
   - Add the SP
   - Rate Schedule is configured as
     - 01-01-2025 → ZZEU1 → Configure Usage Group 1 in Rate Schedule (D2-CCBRateScheduleLookup) extendable lookup.
       - Usage Group 1 with Math Usage Rule

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        - 01-03-2025 → ZZEU2 → Configure Usage Group 2 in Rate Schedule (D2-CCBRateScheduleLookup) extendable lookup.
            - Usage Group 2 with Math Usage Rule
        - 01-05-2025 → ZZEU3 → Configure Usage Group 3 in Rate Schedule (D2-CCBRateScheduleLookup) extendable lookup.
            - Usage Group 3 with Math Usage Rule
    - Ensure interval MC has reads from 01-01-2025 12:00AM to 01-05-2025 12:00AM
4. Create usage transaction for dates
    - Interval Start 01-01-2025 12:00AM
    - Interval End From 01-05-2025 12:00AM
    - Interval End To 01-05-2025 12:00AM
    - Scalar Start 01-01-2025 12:00AM
    - Scalar End From 01-05-2025 12:00AM
    - Scalar End To 01-05-2025 11:59PM
5. UT is created (No Sub UT's are created) with the first effective rate as of UT start date/time with Usage Group 1

Test Case 4: Closing Bill - No Sub UT created when period is less than 24 hours due to rate change, Suppress Partial Day Effective Rates (Y/N) = Y and Use Only One Rate (Y/N) = Y

- Create device and install to an SP
- Create reads. i.e. Interval Reads for 01/01/2025 12:00AM to 01/05/2025 12:00AM
- Create Usage Subscription that uses Usage Subscription Type that derives the Usage Group using rates. Set the US start date as 01/01/2025 12:00AM
    - Setup Rate 1 = 01/01/2025
- Inactivate the Usage Subscription with End Date = 01/05/2025 12:00AM USSP Stop Date = 01/04/2025 12:00AM
- Create a closing UT using Math Rule for
    - Interval Start 01/03/2025 12:00AM
    - Interval End From 01/05/2025 12:00AM
    - Interval End To 01/05/2025 12:00AM

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    - Scalar Start 01/03/2025 12:00AM
    - Scalar End From 01/04/2025 12:00AM
    - Scalar End To 01/04/2025 11:59PM
- Verify that resulting UT is calculated and dates are not shifted
    - UT calculated from 01/03/2025 12:00AM to 01/05/2025 12:00AM
    - Interval Start 01/03/2025 12:00AM
    - Interval End From 01/05/2025 12:00AM
    - Interval End To 01/05/2025 12:00AM
    - Scalar Start 01/03/2025 12:00AM
    - Scalar End From 01/04/2025 12:00AM
    - Scalar End To 01/04/2025 11:59PM
- No Sub UT created and only 1 rate is picked up.

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